

StudyHub

A full-stack AI-powered academic study management platform

Central Asian Unive
Engineering Fa
Final Project Show
2025-

151

GIT COMMITS

50+

PULL REQUESTS

29

API ENDPOINTS

8

DB TABLES

7

DOCKER SERVICES

600+

RAG DOC LINES

OVERVIEW

Problem

Students have no central place for module materials — files scattered across drives, group chats, and tools. Flashcard creation is manual and slow.

- No structure across modules and weeks
- Resources hard to find and share
- No integrated AI study tools

Solution

One platform organised as **Module** → **Run** → **Week** → **Resources** with AI built in at every step.

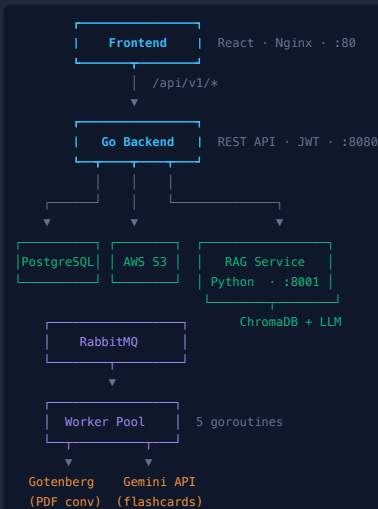
- Upload files and links per week, visible to all classmates
- AI flashcard decks auto-generated from uploaded PDFs
- RAG chatbot for system and study material questions
- Comments and upvotes on weekly resources

Tech Stack

Frontend	React 18, TypeScript, Tailwind, shadcn/ui
Backend	Go, Chi router, PostgreSQL
File Storage	AWS S3 + SHA-256 deduplication
AI Flashcards	Google Gemini 2.5 Flash
RAG Chatbot	Python, FastAPI, LangChain, ChromaDB
Async Queue	RabbitMQ + 5-goroutine worker pool
PDF Convert	Gotenberg
Infra	Docker Compose, GitHub Actions, AWS EC2

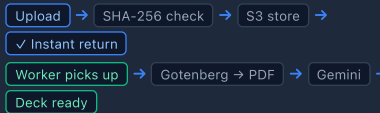
ARCHITECTURE & PIPELINES

System Architecture

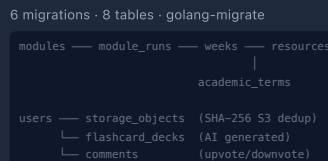


Async Flashcard Pipeline

Upload returns in <1s. Gemini processing (10–30s) happens in the background via RabbitMQ.



Database Design



CI/CD Pipeline

- Path filter** — only changed service rebuilds (saves ~3 min/deploy)
- 4 backend checks** — go test · golangci-lint · gosec · govulncheck
- GHCR images tagged latest + commit SHA for rollback
- Auto SSH deploy into EC2 on every push to main

Future Work

- Auto-index S3 uploads into RAG
- Full-text resource search
- Push notifications for flashcard completion
- Streaming LLM responses (SSE)
- Mobile-responsive UI

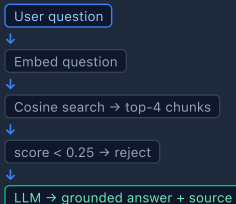
RAG CHATBOT

RAG Chatbot — Build Time vs Runtime

BUILD TIME (ONCE)



RUNTIME (EVERY QUESTION)



Functional Requirements Coverage

- FR-01** Context-aware responses via ChromaDB similarity search
- FR-02** 600-line knowledge base: features, all 29 APIs, auth, architecture
- FR-03** Two-layer rejection — cosine score gate + system prompt instruction
- FR-04** Chat widget on every page with typing indicator and message history
- FR-05** Source attribution — document name and page shown per answer
- FR-06** Multi-turn — last 6 messages as history + retrieval query enrichment

Challenges Solved

S3 Race Condition

Duplicate uploads bypassed hash check simultaneously. Fixed with DB unique constraint on hash — second insert fails gracefully.

RabbitMQ Startup Crash

Go started before RabbitMQ was ready. Fixed with exponential backoff retry loop.

RAG Follow-up Questions

Short follow-ups scored too low in ChromaDB alone. Fixed by prepending previous user message to retrieval query.

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Repo: github.com/karl1mov-u/StudyHub

StudyHub

Go · React · Python · Docker · AWS

Module: 4-S

Central Asian University · Spring 2